Token Parsimony

Toward Semantic Compression Substrates for Evidence-Bearing Artificial Intelligence

Why Future AI Systems Will Spend Less Time Reading and More Time Remembering

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Abstract

Modern large language model systems are operationally inefficient observers.

Most contemporary agent architectures repeatedly ingest the same repositories, logs, transcripts, filesystems, and execution traces across reasoning cycles, reconstructing situational awareness from raw substrate exposure instead of preserving compact semantic continuity. As systems scale, the overwhelming majority of token expenditure shifts away from reasoning itself and toward repeated reacquisition of already-known operational state.

Under hyperscale deployment conditions, redundant substrate reacquisition becomes a primary infrastructure burden. Token parsimony reduces not only inference expenditure, but also the upstream transport, synchronization, hydration, and storage costs associated with repeated semantic reconstruction. The result is reduced semantic input bloat across distributed inference substrates and datacenter-scale orchestration systems — in practical terms, more computationally efficient AI infrastructure.

This paper introduces **Token Parsimony** as a formal architectural principle for semantic systems. Token parsimony is the disciplined minimization of substrate exposure while preserving replay-relevant semantic continuity, inferential sufficiency, and governance integrity.

We argue that the dominant inefficiency in current agent systems is not model inference itself, but repeated semantic reacquisition through raw token consumption. Governed semantic receipts, compact operational summaries, inferentially sufficient transition artifacts, and replay-oriented semantic state transitions can dramatically reduce operational token expenditure while simultaneously increasing auditability, replayability, and epistemic continuity.

Rather than treating tokens as disposable context bandwidth, token parsimony treats them as a constrained thermodynamic resource within semantic reactors.

Under this framing, raw output flooding becomes analogous to energy dissipation in physical systems: computationally expensive, destabilizing, and semantically redundant.

The paper further argues that token parsimony naturally favors constitutional semantic architectures. Systems capable of compact semantic continuity require governed state transitions, semantic custody, replayable operational lineage, and substrate-aware closure guarantees. This leads toward a new class of systems in which agents reason primarily from semantic state transitions rather than repeated raw substrate observation.

The result is a shift from *context accumulation* toward *semantic continuity engineering*.

1. Introduction

The dominant operational pattern in modern AI systems is profoundly wasteful.

An agent receives a task. It searches a repository. It loads files. It scans logs. It reconstructs operational state. Then, in the next session, another agent performs nearly the same sequence again. The substrate may have changed only minimally, but the semantic acquisition process effectively restarts from zero.

As repositories, operational corpora and histories grow, token expenditure increasingly shifts away from reasoning itself and toward repeated reacquisition of already-known substrate state. Systems repeatedly pay the observational cost of rebuilding situational awareness.

The problem is not merely economic.

Repeated raw observation produces:

- context flooding,
- replay instability,
- epistemic drift,
- semantic inconsistency,
- audit discontinuity,
- hallucination opportunities,
- and degraded operational coherence.

Current architectures implicitly assume that intelligence consists primarily of repeatedly observing raw substrate.

This paper argues the opposite.

Scalable intelligence increasingly depends upon preserving compact semantic continuity across operations while minimizing repeated substrate exposure.

This principle is termed **Token Parsimony**.

2. Formal Model

Token parsimony can be described using a minimal operational formalism.

Let:

- S_t denote substrate state at time t
- $O(S_t)$ denote raw substrate observation
- R_t denote a semantic receipt
- T denote a semantic transition operator
- $C(x)$ denote token cost of representation x

Let:

$$S_t^{\mathcal{Q}} \subseteq S_t$$

represent the minimal substrate subset required for task class $\mathcal{Q}$.

Define:

$$T : (S_{t-1}, S_t) \to R_t$$

The transition operator T encodes both the state mutation and its receipt projection.

Receipt generation must be deterministic with respect to declared normalization, environment, and capture semantics.

A semantic receipt is therefore defined as:

$$R_t = T(S_{t-1}, S_t)$$

A receipt is valid iff it preserves inferential sufficiency over $S_t^{\mathcal{Q}}$.

Define the Parsimony Ratio:

$$P = \frac{\sum_{i=1}^{n} C(O(S_i))}{\sum_{i=1}^{n} C(R_i)}$$

where:

- numerator = cumulative token cost of raw substrate observation,
- denominator = cumulative token cost of semantic continuity artifacts.

Parsimony is meaningful over operational horizons rather than single transitions.

A parsimonious system therefore seeks to maximize P subject to inferential sufficiency and bounded escalation constraints.

The objective is not arbitrary compression, but preservation of operationally necessary meaning with minimal substrate exposure.

3. Inferential Sufficiency

The central operational question of token parsimony is:

> "What is the minimum semantic artifact required to preserve future inferential validity?"

This leads to the concept of **inferential sufficiency**.

Let:

$$Answer(q \mid x)$$

denote the system's output for task q given representation x, including decisions, generated artifacts, or execution outcomes.

A receipt R is inferentially sufficient for task class $\mathcal{Q}$ iff:

$$\forall q \in \mathcal{Q}, \quad Answer(q \mid R) = Answer(q \mid S)$$

or triggers bounded escalation.

Equality may be exact or defined over a task-specific equivalence class (e.g., identical decisions, outputs, or bounded divergence).

Bounded escalation means:

- the system detects insufficiency,
- selectively reacquires substrate,
- and limits additional token expenditure to a bounded retrieval scope.

Formally:

$$C(\text{escalation}) \leq \epsilon \cdot C(O(S_t))$$

for some policy-defined:

$$\epsilon \in (0, 1]$$

Escalation policies therefore constrain worst-case token reacquisition relative to full substrate observation.

Escalation retrieves additional portions of S_t not present in the current representation when required for $\mathcal{Q}$.

Token parsimony does not eliminate substrate access.

It governs escalation.

The retrieval hierarchy becomes:

1. semantic receipt,
2. compact operational summary,
3. structured fetch,
4. partial substrate exposure,
5. full substrate exposure.

Most current systems default immediately to level 5.

Parsimonious systems attempt to remain at the lowest admissible level.

4. Receipts as Semantic State Transitions

The central insight of token parsimony is that most operations do not require reproduction of raw output.

They require preservation of semantic state transitions.

For example, a conventional write operation may emit:

- the full file,
- surrounding context,
- timestamps,
- path metadata,
- and verbose explanation.

A parsimonious system instead emits:

```
write:engine.py:hashA->hashB:+14/-2
```

This compact representation preserves:

- object identity,
- mutation direction,
- mutation magnitude,
- replay relevance,
- and continuity lineage (the chain of receipts linking prior and current state).

Similarly:

```
run:smoke_test:exit0:stdout_
```

```
hash manifest:src:241files:manifest_
hash audit:98receipts:0open:3warnings
```

These are not merely compressed outputs.

They are governed semantic transition vectors.

5. Receipt Taxonomy

Receipts naturally partition into distinct operational classes.

5.1 State Receipts

Represent substrate snapshots or continuity anchors.

Examples:

- manifests,
- directory snapshots,
- inventory hashes.

5.2 Transition Receipts

Represent substrate mutations.

Examples:

- writes,
- diffs,
- merges,
- state transitions.

5.3 Execution Receipts

Represent runtime operations.

Examples:

- command execution,
- exit codes,
- stdout hashes,
- elapsed time.

5.4 Claim Receipts

Represent semantic assertions bound to evidence.

Examples:

- audits,
- investigations,
- governed claims,
- refutations.

This taxonomy becomes increasingly important as systems transition from observational architectures toward semantic operating environments.

6. End-to-End Example

Consider a conventional CI pipeline.

A standard agentic workflow may repeatedly:

- scan the repository,

- load changed files,
- read full build logs,
- inspect test outputs,
- and reconstruct prior operational state.

Suppose:

- repository size = 40 MB,
- build log = 300k tokens,
- test output = 80k tokens,
- repeated across multiple agents.

Operational token cost rapidly becomes enormous.

A parsimonious pipeline instead operates primarily through receipts.

Step 1 — Repository Continuity

Instead of rescanning the repository:

```
manifest:src:241files:manifest_hash
```

If unchanged, no escalation occurs.

Step 2 — Build Execution

Instead of ingesting full logs:

```
run:build:exit0:stdout_hash
```

Only failures trigger escalation.

Step 3 — Test Verification

```
run:test_suite:exit1:stdout_hash
```

Only the failing test shard is fetched during escalation (e.g., 2k tokens instead of 80k total output).

Step 4 — Drift Analysis

```
write:parser.py:hashA->hashB:+6/-1
```

Agents infer likely affected systems without reacquiring unrelated substrate.

The result is a receipt-dominant execution regime in which substrate exposure becomes selective rather than default.

Preliminary architectural analysis suggests reductions on the order of 1–3 magnitudes under receipt-dominant operational patterns.

7. Hash Semantics

Token parsimony relies heavily on compact identity primitives such as hashes.

However, hashes require careful semantic framing.

A hash is not evidence.

A hash is a governed pointer to evidence.

Systems must explicitly define:

- what is hashed,
- normalization rules,
- collision assumptions,
- environmental stability guarantees,
- and capture semantics.

Examples include:

- raw content hashes,
- normalized AST hashes,
- execution trace hashes,
- or manifest hashes.

The following invariant must hold:

$$hash(x) = hash(y) \Rightarrow x \equiv y$$

under the declared normalization regime.

Equivalence is therefore defined by the hashing regime rather than raw byte identity.

Execution hashes additionally require declared capture semantics:

- full stdout,
- filtered output,
- deterministic projections,
- environment normalization,
- and ordering guarantees.

Adversarial or mis-specified hashing regimes can produce false equivalence; governance must therefore bind hashes to declared construction procedures.

Failure to define hashing semantics creates false continuity risks.

Two receipts with identical hashes imply continuity only within the declared hashing jurisdiction.

8. The Thermodynamics of Context

Current AI systems behave thermodynamically poorly.

They repeatedly dissipate computational energy through redundant substrate observation.

Within the token parsimony model:

- energy corresponds to token expenditure,
- entropy corresponds to semantic uncertainty or loss,
- dissipation corresponds to redundant reacquisition of already-known substrate.

Large context windows partially mask this inefficiency but do not solve it. Increasing context merely enlarges the reactor chamber while preserving inefficient operational physics.

Under token parsimony:

- receipts replace repeated file reads,
- semantic transitions replace repeated state reconstruction,
- governed hashes replace repeated payload transmission,
- and inferential references replace observational flooding.

The shift resembles the emergence of symbolic accounting systems within human civilization: compact semantic structures replace repeated physical enumeration.

9. Governance and Semantic Closure

Token parsimony cannot operate reliably without governance.

A compact receipt is only useful if its semantic continuity is trusted.

This creates a direct dependency between:

- token parsimony,
- semantic custody,
- replay integrity,
- governance closure,
- and constitutional continuity.

A receipt lacking:

- provenance,
- admissibility,
- operational lineage,
- or replay guarantees

cannot safely replace substrate observation.

Receipts are assumed to reside in a durable, queryable store with append-only or governed mutation semantics.

Receipt stores must preserve ordering or causal linkage sufficient to reconstruct transition lineage.

Thus semantically parsimonious systems naturally favor governed semantic architectures.

As semantic artifacts become smaller, the operational meaning carried by each artifact increases dramatically.

This creates governance-compressed semantic state: small symbolic structures carrying disproportionately large operational authority.

10. Failure Modes

Token parsimony introduces new architectural risks.

These include:

- semantic oversummarization,
- inferential insufficiency,
- phantom continuity,
- governance spoofing,
- substrate drift,
- and illegitimate semantic condensation.

Consider a failure example.

A manifest receipt reports:

```
manifest:src:241files:manifest_hash
```

A hidden deployment artifact was excluded from the hashing scope due to a configuration bug.

Future reasoning assumes repository continuity and deploys stale infrastructure.

The failure was not cryptographic.

It was semantic.

The continuity claim exceeded the actual hashing jurisdiction.

This illustrates a central principle:

> Semantic continuity claims are bounded by declared substrate scope.

Governance systems must therefore validate:

- receipt completeness,
- hashing jurisdiction,
- escalation correctness,
- continuity admissibility,
- and inferential scope.

11. Escalation Policy

Token parsimony depends upon governed escalation.

Receipts are not universally trusted. They are conditionally sufficient.

An escalation policy determines:

- when semantic continuity is insufficient,
- which retrieval layer to escalate toward,
- maximum allowable escalation cost,
- replay admissibility requirements,
- and recovery behavior.

Upon escalation failure, systems may invalidate dependent receipts and revert to a higher observation tier.

Escalation transforms token parsimony from blind compression into adaptive semantic governance.

Without escalation policy, parsimonious systems become brittle.

With governed escalation, they become operationally scalable.

12. Substrate Liveness

A governance-complete semantic system can still fail operationally if its substrate loses liveness.

This paper defines **substrate liveness** as the operational viability of the substrate required to sustain semantic state transitions.

Examples include:

- lock contention,
- stalled subprocesses,
- deadlocked receipt stores,
- WAL starvation,
- exhausted descriptors,
- and recursive governance contention.

Substrate liveness is orthogonal to semantic legitimacy.

A constitutionally valid system may still become operationally frozen.

This becomes especially important in recursive governance systems where:

- audit systems audit themselves,
- receipt stores recursively inspect receipts,
- and governance substrates recursively govern their own continuity.

Semantic reactors therefore require not only constitutional legitimacy, but operational thermodynamics.

13. Toward Semantic Operating Systems

Token parsimony ultimately implies a new class of operating environment.

Conventional systems are:

- process-centric,
- file-centric,
- or message-centric.

Semantically parsimonious systems become:

- transition-centric,
- receipt-centric,
- and continuity-centric.

Meaning increasingly resides not in repeated substrate exposure, but in governed semantic transitions.

Agents stop repeatedly rediscovering operational state and instead inherit compact semantic continuity through replayable receipt structures.

This transforms semantic systems from observational machines into continuity-preserving semantic civilizations.

These properties suggest a minimal execution model in which agents operate over receipt graphs rather than raw substrate.

14. Conclusion

The future scalability limits of AI systems are unlikely to emerge primarily from model intelligence.

They are more likely to emerge from semantic inefficiency.

Current architectures repeatedly spend enormous computational resources reacquiring operational state that could instead persist as governed semantic continuity.

Token parsimony reframes the problem fundamentally.

The question is no longer:

> "How large can context windows become?"

but:

> "How little substrate exposure is actually necessary?"

This inversion changes the architecture of intelligent systems profoundly.

It favors:

- semantic continuity,
- inferential sufficiency,

- governance closure,
- replayable operations,
- compact semantic transitions,
- and substrate-aware escalation.

In such systems, intelligence increasingly shifts away from repeated observation and toward continuity-preserving semantic operation.

The result is not merely cheaper computation.

It is the emergence of systems capable of reasoning through governed semantic continuity rather than repeated reacquisition of already-known state.

Corpus-Wide Reference Positioning

This paper argues that modern large language model systems are operationally inefficient observers. Its nearest external comparison fields are token, semantic, ai. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Statistical compression can improve efficiency while destroying distinctions needed for interpretation; semantic labels can conversely overstate stability that usage does not support. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Retrieval, tool use, and policy layers can reduce unsupported output without eliminating model error, source error, omission, bias, or the environmental and institutional costs of scale. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant

work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Rico Sennrich, Barry Haddow, and Alexandra Birch. "Neural Machine Translation of Rare Words with Subword Units." *ACL 2016*, pp. 1715-1725. DOI: 10.18653/v1/P16-1162. https://doi.org/10.18653/v1/P16-1162 **Relationship:** Predecessor subword tokenization method used as a contrast to semantically governed token boundaries.

[**E2**] Taku Kudo and John Richardson. "SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing." *EMNLP 2018: System Demonstrations*. DOI: 10.18653/v1/D18-2012. https://doi.org/10.18653/v1/D18-2012 **Relationship:** Predecessor tokenizer implementation used to separate statistical segmentation from semantic admissibility.

[**E3**] Peter D. Grunwald. *The Minimum Description Length Principle*. MIT Press, 2007. mitpress.mit.edu source **Relationship:** Predecessor theory of compression and model selection; compression is not assumed to preserve governance or meaning.

[**E4**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E5**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E6**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E7**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

Ableman corpus

[**A1**] Adam Ableman. *Governed Semantic Tokens: Tokenization as an Admissibility Boundary.* Ableman Corpus current edition; work identity `PAPER.PUB.18469507`;
source SHA-256
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Legends.* Ableman Corpus current edition; work identity `PAPER.PUB.20330324`;
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[**A3**] Adam Ableman. *Governed Controlled Natural Language (GCNL).* Ableman Corpus current edition; work identity
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[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
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[**A5**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
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[**A7**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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